

Forward algo: $\underline{O} = [O_1, \dots, O_T]$

$P(\underline{O} = \underline{o} \mid \text{HMM model})$ Prob. obs. seq. $\underline{o}$ given model

More efficient than brute force

Viterbi:

given obs. $\underline{o}$ and HMM Model what is most likely State Seq.

find seq. $\underline{q} = [q_1, \dots, q_T]$ that max $P(X_1 = q_1, \dots, X_T = q_T \mid \underline{O} = \underline{o})$

Baum-Welch = find optimal parameters for model given obs. seq. and knowledge of state space and obs. symbols

Congestion detectors

Mult. Faults $\rightarrow$ 2 fault types: Poor wireless link & Congestion in network

- Modelled as 4 State Markov Chain and obs. RTT & RSSI

then

- Obs. Seq. $\rightarrow$ HMM + viterbi $\rightarrow$ Est. Fault State Seq.